Whiteness Training for Colorless Feedback Delay Networks

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Code: github.com/snlehton/whiteness-fdn

Abstract

A homogeneous feedback delay network (FDN) trained on the whiteness of its impulse response, plus three terms that shape its first 100 ms, comes within −1.1 to +0.5 dB of DiffFDN's [1] modal spread, with a smaller largest echo in all six of its configurations, at a small fraction of its training cost. We unroll the FDN in time and minimize the normalized off-zero autocorrelation of the impulse response: the sum of $r[k]^2$ over all lags $k \neq 0$, divided by $r[0]^2$. It equals the normalized variance of $|H|^2$, and its gradients do not vanish as the response gets longer. Whiteness alone is gamed by near-permutation matrices and sparse responses, so we add a mixing penalty and an early-density term.

Trained gains first collapse onto one delay line. Tapping a single line turns the FDN exactly into one delay line with an allpass in its loop, and its unmixed first echo dominates the onset. Three terms on the first 100 ms remove that echo with the gains left free: the effective number of echoes in windows that hold a fixed number of arrivals, a build-up ramp, and a ceiling on each line's direct echo that rises with its delay. Training takes 16–86 s on a CPU, against 3.5–16 min on a GPU for DiffFDN.

1. Introduction

For a given set of delays, an FDN's coloration is set by its feedback matrix U and its gains b and c. DiffFDN [1] learns all three by sampling H(z) on the unit circle. A spectral-flatness loss on |H| is combined with an ℓ_2/ℓ_1 term that keeps the impulse response dense. This note reports what we found when we reproduced it and then tried a time-domain alternative:

1. Running the authors' code unmodified reproduces the paper's result, and shows three details that matter for building on it (Section 2).
2. The whiteness of the unrolled impulse response is a usable training objective with well-behaved gradients. It needs two counterweights, against permutation matrices and against sparse early responses (Section 3).
3. An exact identity for single-line gains explains why the trained b and c concentrate on one line, and why that produces a loud first echo (Section 4).
4. What we tried first: gains on per-pair unit circles, with b 90° ahead of c, bound every first echo by construction, at the cost of constraining the gains (Section 5).
5. Terms that shape the first 100 ms directly do better with free gains. They count echoes over windows sized by the number of arrivals rather than by time, hold the onset to a build-up ramp, and cap each line's direct echo. The largest echo then ends below DiffFDN's in every configuration (Sections 6 and 7).

Throughout, $H(z) = c^T [D(z)^{-1} - U\Gamma]^{-1} b$ with $D(z) = \mathrm{diag}(z^{-m_i})$, orthogonal U, $\Gamma = \mathrm{diag}(\gamma^{m_i})$ and $\gamma = 0.9999$ during training, at fs = 48 kHz. The configurations are the six of [1, Table 1]: N = 4, 6 and 8 delay lines with two delay sets each. Set #2 clusters short, nearly equal delays and has fewer modes, for example 6640 against 8944 at N = 8.

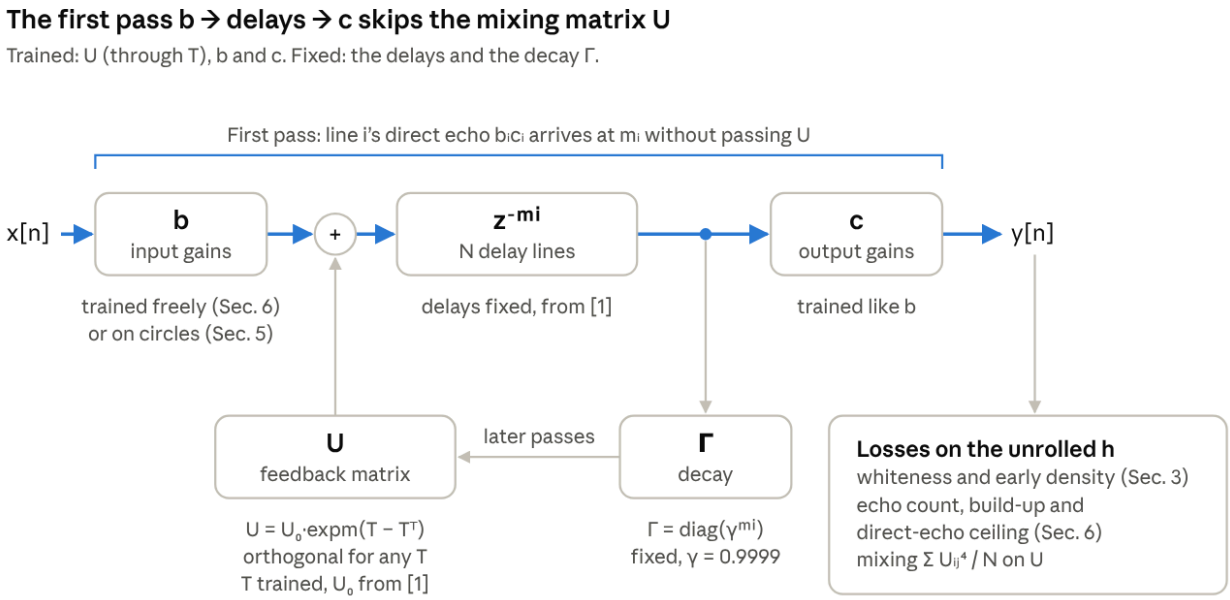


Figure 1. The FDN in vector form. The first trip through each line, $b_i \to z^{-m_i} \to c_i$, never passes U and arrives as one isolated echo at m_i (Sections 4 and 6). Every later echo has been mixed by U at least once.

2. Reproducing DiffFDN

The optimization reproduces. We ran the authors' `dafx23` branch (commit d4dc2f25) unmodified on all six configurations, with the default $\alpha = 300$, 15 epochs and seed 130798. Every run narrows the modal-excitation spread from $\sigma = 5.9$–7.8 dB to 3.4–4.4 dB, and lowers the loudest mode by 4.7–9 dB. The paper's Fig. 6 reports about 3 dB. The results fall in the range of the five parameter sets published in the repository. They also beat typical random orthogonal matrices: 20 Haar draws per delay set give $\sigma = 7.0$–7.5 dB. Early echo density rises too, above Householder and Haar in every configuration.

Modal excitation was computed with a Python reimplementation of the modal decomposition [2]. Pole counting uses the eigenphases of $\mathrm{diag}(\exp(-j\omega m_i))\cdot U$, which need only an N×N eigendecomposition. Against fdnToolbox [3] on the authors' `pretrained00` set, pole angles agree to 1e−10 rad and residue magnitudes to below 0.001 dB. A run takes 3 s, against 122 s for a dense M×M eigendecomposition.

Three details matter when building on the method:

1. **The initialization is close to the identity.** `weights_init_normal` draws W with standard deviation 1/N, not variance 1/N. The matrix $\exp(W - W^T)$ is then weakly coupled: $\|\mathrm{diag}\, U\|^2/N = 0.77$–$0.86$, against 0.06–0.27 for a Haar draw. The RO condition of the listening test in [1] is therefore comb-like, although its modal spread is typical of random matrices.
2. **Modal spread alone can be gamed.** A diagonal matrix (parallel combs, the test's anchor) has the narrowest distribution of all, $\sigma \approx 1.6$ dB, because each comb excites its modes equally. σ is only meaningful together with a mixing or density measure.
3. **The code's loss differs from eq. (17) in two places.** It adds the asymmetric loss of the summed response, weighted by $\sqrt{(\text{batch size})} = 2$. It also leaves the zero-padded frequency bins unmasked. At $z = 0$ each channel evaluates to the constant $c_i(-(U\Gamma)^{-1}b)_i$, so 0–20% of each sample's bins pull $|c_i(\Gamma^{-1}U^T b)_i|$ towards 1, a hidden gain constraint. A StepLR schedule also divides the learning rate by 10 after epoch 10, which explains the knee in the loss curves.

3. A whiteness loss on the unrolled impulse response

A colorless tail should look like white noise, so we drive the off-zero autocorrelation r[k] of the impulse response h towards zero. By Wiener–Khinchin and Parseval, the normalized sum is a spectral-flatness measure on $|H|^2$. It belongs to the same family as the loss of [1], but is computed in the time domain:

$$\mathcal{W}(h) = \frac{\sum_{k\neq 0} r[k]^2}{r[0]^2} = \frac{\mathrm{Var}\ |H(e^{j\omega})|^2}{\left(\mathrm{E}\ |H(e^{j\omega})|^2\right)^2}$$

Gaussian noise scores about 1 and an impulse or an allpass scores 0, so values below 1 mean a sparse response. For a lossless FDN and a window of L samples much longer than the order M, the measure becomes the fourth moment of the modal excitation:

$$\mathcal{W}(h) + 1 \ \to\ \frac{2L}{3}\,\frac{\sum_i |\rho_i|^4}{\left(\sum_i |\rho_i|^2\right)^2}$$

This agrees within 1% from L = 1 s to 40 s on N8 set #1, for both the init and DiffFDN. Used as a metric on 2.5 s responses at γ = 0.9999, DiffFDN has the lowest whiteness of all the FDNs tested, in every configuration (0.94–2.31). The other candidates were the RO init, Householder, Haar and a comb bank.

Unrolled network and gradients. We compute h in PyTorch as a block recursion over the shortest delay, a linear network of depth L / min(m). At N8 set #1 that is 594 blocks for 10 s. It matches direct simulation to 1.7e−17, and a forward and backward pass takes 0.4 s on a CPU. The gradients do not vanish:

- $\|\nabla\mathcal{W}\| / \mathcal{W}$ stays at 1.1–1.2 for windows from 0.5 s to 10 s.
- The gradient direction converges with length: the cosine between the 5 s and 10 s gradients is 0.99 (lossless).
- The sensitivity of h to a rotation of U grows linearly with time (log–log slope 1.00). Because whiteness ignores phase, the loss along random rotation paths stays smooth, with no local minima at 1 s or 10 s.
- At γ = 0.9999, nothing past about 1 s changes the loss, so γ sets the effective window. We train on 2 s.

U is parametrized as $U_0 \cdot \mathrm{expm}(T - T^T)$, with T strictly upper triangular, starting from the init U_0 of [1]. We train with Adam, learning rate 0.01, for 400 steps.

Two counterweights. Whiteness alone is gamed in every configuration. U moves close to a permutation ($\Sigma U_{ij}^4/N$ = 0.67–0.93), which chains the lines into one long loop. Its modes are almost equally excited (σ down to 0.2 dB), but the response is a sparse echo train, with early echo density 0.02–0.18 for N ≤ 6. A permutation also fools $\|\mathrm{diag}\ U\|^2/N$, so the mixing penalty is the peakedness $\Sigma U_{ij}^4/N$. That is 1 for a permutation and 1/N for a Hadamard matrix. Whiteness also rewards a strong early tap, since an impulse is white. An early-sparsity term $\mathcal{S}$ counters this. It averages over 20 ms Hann windows w (10 ms hop), from the first arrival to 0.5 s. The ratio of mean absolute value to RMS is normalized by its Gaussian value $\sqrt{(2/\pi)}$, so the term is 0 for noise and approaches 1 for isolated echoes. The full loss is:

$$\mathcal{L} = \mathcal{W}(h) + \frac{1}{N}\sum_{i,j} U_{ij}^4 + 3\,\mathcal{S}(h), \qquad \mathcal{S}(h) = \operatorname*{mean}_w \left(1 - \frac{\mathrm{mean}\,|h_w| \,/\, \mathrm{rms}(h_w)}{\sqrt{2/\pi}}\right)$$

A density weight of 1 gives better coloration but lower early density than DiffFDN. At weight 3, density reaches DiffFDN's at N = 6 and 8, and at weight 10, σ returns to the Haar level (6.0 dB on N6 set #1).

4. Why trained gains collapse onto one line

When b and c are trained freely with the loss above, both concentrate on one or two lines, about 1.5 lines each against 2–5 for DiffFDN. They tend to pick the same line. On N4 set #1 the first echo, one pass through line 1 at 31 ms, then carries 34% of the impulse response's energy at T60 = 2.5 s. Without the density term it carries 43%. The reason is an exact identity for single-line gains.

Let R be the other N − 1 lines. With b = c = e_1, line 1 sees the rest of the network as a filter A(z) from its output back to its input:

$$H(z) = \frac{1}{z^{m_1} - A(z)}, \qquad A(z) = U_{11} + U_{1R}\left(D_R(z)^{-1} - U_{RR}\right)^{-1} U_{R1}$$

Because U is orthogonal, A is a lossless scattering of the other lines, which makes it an allpass, as in allpass FDNs [4]. In the lossless case each mode's amplitude then follows from the group delay of that allpass at the mode frequency:

$$|\rho_k| = \frac{1}{m_1 + \tau_A(\omega_k)}$$

Numerically, on N6 set #1 with a Haar U, |A| = 1 to six decimals and the formula matches the residues to 1e−13 relative error. A flat modal excitation therefore needs a flat group delay of the rest of the network. Energy has to return to line 1 quickly instead of ringing among the other lines. This is the classic delay loop with an allpass diffuser inside, rediscovered by the optimizer.

It is not automatic. For random matrices, single-line gains are the worst layout (modal σ in dB, mean of 3 Haar draws):

Configuration	b = c = e_i (one line)	b = e_i, c = e_j (two lines)	b = c = 1
N4 set #1	8.78	5.53	5.92
N4 set #2	9.18	5.79	6.20
N6 set #1	9.28	5.94	6.17
N6 set #2	9.31	5.97	6.25
N8 set #1	10.70	7.18	6.95
N8 set #2	10.77	7.26	7.03

The trained U builds the flat allpass around the chosen line. The price is the first trip around that line, which passes no mixing, so it arrives as one loud echo, and the other lines join only one trip later. Spreading the gains has its own cost. Each mode then reaches the output through several lines with mode-dependent phases. The sum interferes like speckle, and the spread of $|c^T v|^2$ adds about 5.6 dB. On the trained N6 set #2 result, spreading c evenly over all lines (same U and b) raises σ from 1.40 to 5.72 dB, while early echo density only moves from 0.47 to 0.50.

5. What we tried first: gains on pair circles

Before shaping the transient directly, we tried to bound it by construction. Fixed b = c = 1 lets every line contribute with no first echo standing out, while trained gains use their freedom against interference. To keep some of both, each pair of lines (2k, 2k+1) shares one circle, with one trained angle θ_k per pair and b a quarter turn ahead of c:

$$\begin{pmatrix} c_{2k} \\ c_{2k+1} \end{pmatrix} = \sqrt{2}\begin{pmatrix} \cos\theta_k \\ \sin\theta_k \end{pmatrix}, \qquad \begin{pmatrix} b_{2k} \\ b_{2k+1} \end{pmatrix} = \sqrt{2}\begin{pmatrix} -\sin\theta_k \\ \cos\theta_k \end{pmatrix}, \qquad k = 0, \ldots, \tfrac{N}{2} - 1$$

What this guarantees:

- **Fixed power, as for b = c = 1.** $\|b\|^2 = \|c\|^2 = N$, and every pair carries the same power. At least $N/2$ lines carry each vector, and no gain exceeds $\sqrt{2}$.
- **Bounded, balanced first echoes.** The first pass through the two lines of a pair gives $b_i c_i = \mp \sin 2\theta_k$. Both have the same size, and neither exceeds 1, the b = c = 1 level. When c concentrates on one line of the pair, b moves to the other and that pair's first echo goes silent instead of loud. A line with a large c always gets a small b.
- **Few, smooth parameters.** $N/2$ angles in total, and signs change continuously through the angle.

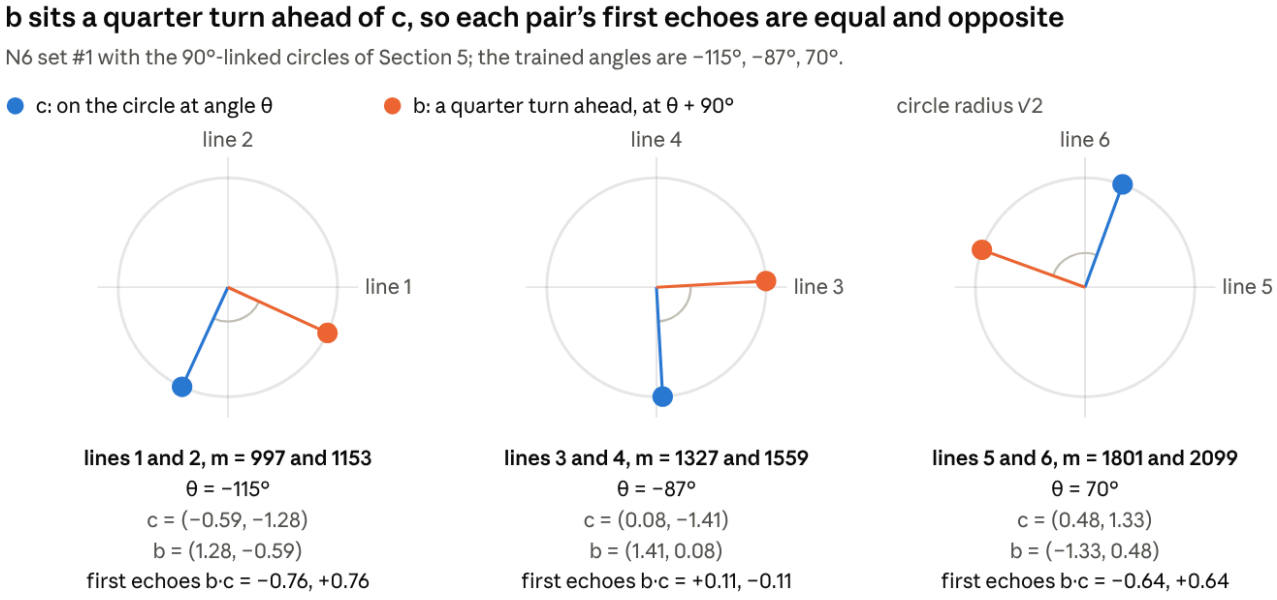


Figure 2. The three pair circles of a trained N6 set #1 result. Each pair settles at its own angle, and the pair's two first echoes come out equal in size and opposite in sign, never above 1.

The angles start uniformly at random, and pairs are adjacent lines of the sorted delay list. Two simpler variants were tested as ablations. The first gives b and c independent angles on the same circles (N angles). It reaches the same coloration at N = 8, but at N = 4 each pair can collapse onto one line. On N4 set #2 line 4 then gets b·c = −1.89, a single echo with 17% of the energy. The second puts only c on the circles (b = 1) and lands between b = c = 1 and the independent circles. Sharing one angle between b and c would not help, since b = c allows b·c = 2 on one line. The link works, but it fixes how each pair's b and c relate. With the transient trained directly (Section 6), free gains do better and the circles are not needed.

6. Shaping the transient

The pair circles constrain the gains to avoid a loud onset. Training the first 100 ms directly does better, with b and c left free. Three terms do it: the effective number of echoes per window, a build-up ramp, and a ceiling on each line's direct echo. All measure $\tilde{h}[n] = h[n] \cdot \gamma^{-n}$, the response with its decay removed (exact for a homogeneous FDN), against the mean power P of its tail from 0.5 s on.

Per-sample level targets fail. The first attempt held every early sample to the tail level. With $(|\tilde{h}|/\sqrt{P} - 1)^2$, a silent sample costs 1 but an echo ten times too loud costs 81. Training then bought a quiet first 100 ms and moved the burst just past the window. A symmetric ±10 dB band in dB closes that loophole but barely trains, because most early samples are reached by no path yet and add only a constant.

The echo budget. Between arrivals the output is exactly zero. An echo arrives at every $n = \Sigma\, k_i m_i$, where $k_i \geq 0$ counts the passes through line i in any order. The number of such k for a given n is the denumerant of n, which for coprime delays grows as $n^{N-1} / ((N-1)! \cdot \Pi m_i)$ [7, 8]. While different k rarely share a delay, summing this over $n \leq t$ counts the arrivals. Shifting t by half the sum of the delays is our own first-order correction, of the kind that appears in classical bounds on the denumerant [8]. It stays within about 20% of the exact counts below until the response turns dense. Echo density in FDNs was analysed in [6]:

$$N_{\text{arr}}(t) \approx \frac{\left(t + \frac{1}{2}\sum_i m_i\right)^N}{N!\,\prod_i m_i} - 1, \qquad \frac{dN_{\text{arr}}}{dt} \approx \frac{t^{N-1}}{(N-1)!\,\prod_i m_i}$$

Configuration	arrivals in the first 50 ms	in the first 100 ms	about one per sample after
N4 set #1	3	12	1 s
N4 set #2	7	35	0.6 s
N6 set #1	10	78	0.3 s
N6 set #2	12	122	0.3 s
N8 set #1	33	465	0.13–0.2 s
N8 set #2	566	2876	0.05–0.1 s

A lossless loop delivers energy at roughly the tail's rate from the first echo on. On N4 set #1, three echoes share the first 20 ms, so each sits about 25 dB above the tail even if they share the energy evenly.

Effective echo count. In a window w holding n arrivals, $K = (\Sigma\, \tilde{h}^2)^2 / \Sigma\, \tilde{h}^4$ is the effective number of echoes: n when they are equally loud, 1 for a lone echo. It is scale-free per window, so going quiet buys nothing. As a fourth moment, it still penalizes one loud echo among quiet ones, where the ℓ_1/ℓ_2 term $\mathcal{S}$ has saturated:

$$\mathcal{E}(h) = \operatorname*{mean}_w \log \frac{n_w}{K_w}, \qquad K_w = \frac{\left(\sum_{n\in w} \tilde{h}[n]^2\right)^2}{\sum_{n\in w} \tilde{h}[n]^4}$$

Each window spans 8 arrivals or 20 ms, whichever is longer, and the next starts halfway through, up to 100 ms. That is 50 ms at the start of N4 set #1 and 20 ms throughout on N8 set #2. Fixed 20 ms windows fail at N = 4, because they hold 1–3 arrivals each. The term then rated an N4 set #1 model with one echo 36 dB above the tail better than DiffFDN (4.85 against 4.95). A window must also not start exactly at the first echo, where its Hann taper is zero. At N = 8, fixed 20 ms windows starting 10 ms early work slightly better.

Build-up. DiffFDN starts 2–6 dB below its tail and reaches it after 150–200 ms, while a model trained on $\mathcal{E}$ alone starts at full level. $\mathcal{B}$ holds each window's level $\ell = 10 \log_{10}(\text{mean}\, \tilde{h}^2 / P)$, over windows up to 0.3 s, to a ramp. The ramp rises from −6 dB at the first arrival t_0 to 0 dB at 150 ms and stays there, so an overshoot after the rise costs too:

$$\mathcal{B}(h) = \operatorname*{mean}_w \left(\frac{\ell_w - \min\left(0,\ -6 + 6\,\frac{t_w - t_0}{150\ \text{ms}}\right)}{10} \right)^2$$

Direct echoes. The path input → line i → output never passes U. It arrives at m_i with amplitude $b_i c_i$. DiffFDN's direct echoes grow with delay. On N6 set #2 its three short lines sit +10 to +19 dB above the tail and its long ones +15 to +22 dB; on N4 set #1 the shortest sits at +4 dB. Each product $b_i c_i$ sets one early echo, while the tail's modal excitation depends on b and c through the eigenvectors of U. $\mathcal{D}$ penalizes each direct echo above a ceiling that rises linearly with delay, from +10 dB for the shortest line to +22 dB for the longest:

$$\mathcal{D} = \operatorname*{mean}_i \left(\frac{\left[10 \log_{10} \frac{(b_i c_i)^2}{P} - 10 - 12 \, \frac{m_i - m_{\min}}{m_{\max} - m_{\min}} \right]_+}{10} \right)^2$$

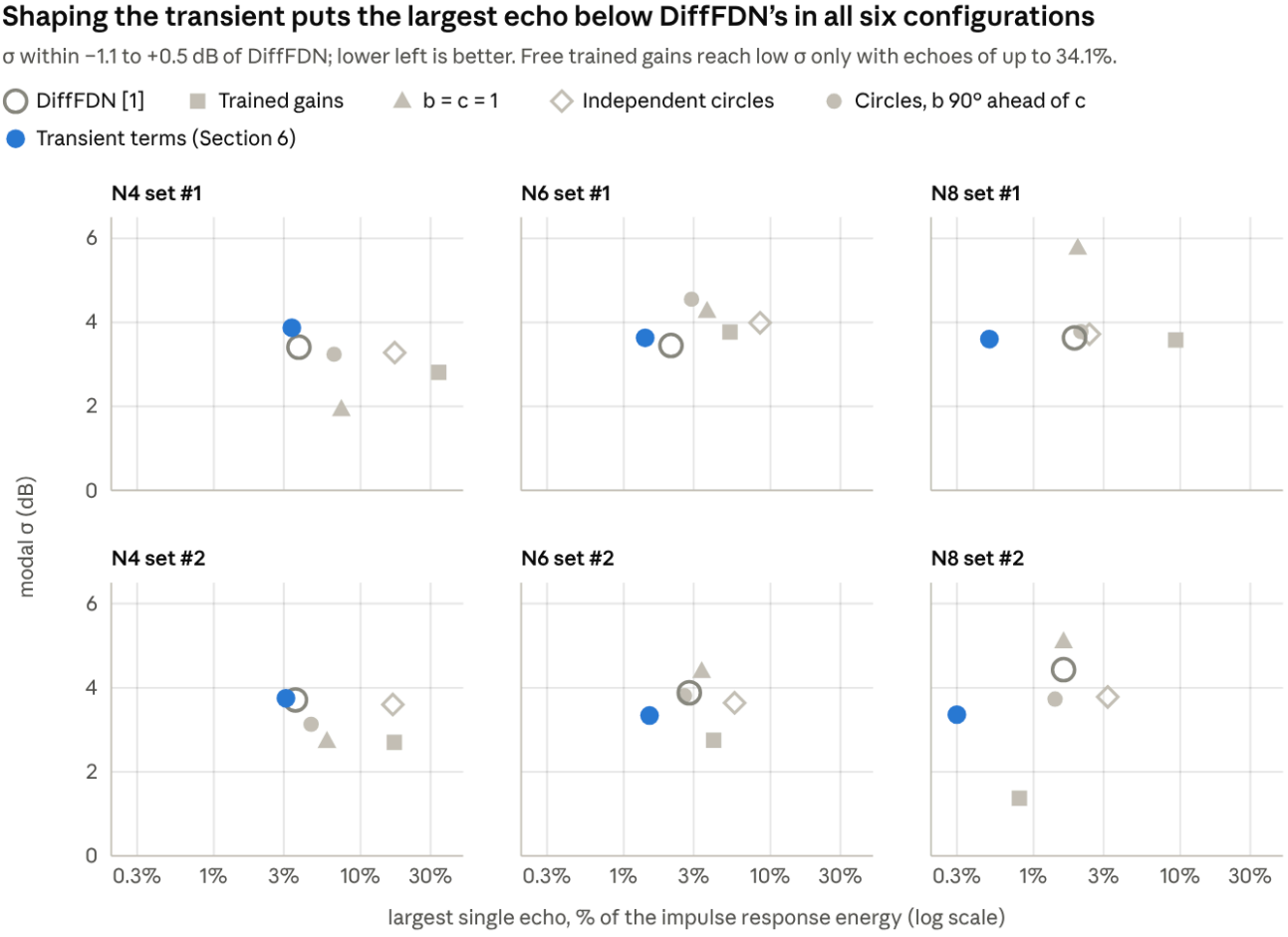
Objectives used. With b and c trained freely, from the init of [1]:

$$\mathcal{L}_{N=4,6} = \mathcal{L} + 0.5\,\mathcal{E} + 30\,\mathcal{B} + 3\,\mathcal{D}, \qquad \mathcal{L}_{N=8} = \mathcal{L} + 0.5\,\mathcal{E}_{20\,\mathrm{ms}}$$

At N = 8, $\mathcal{B}$ also builds up but costs coloration (σ 3.94 against 3.60 dB on N8 set #1), so it is left out there.

7. Results

With the transient terms of Section 6 and free gains, the largest single echo is smaller than DiffFDN's in all six configurations. Modal σ lies within −1.1 to +0.5 dB of DiffFDN's: lower on N6 set #2 and N8 set #2, within 0.05 dB on N4 set #2 and N8 set #1, and higher on N6 set #1 (+0.18) and N4 set #1 (+0.46). The 90°-linked circles of Section 5, and every other gain option, now sit to the right of these results (Figure 3). Training takes 16–86 s per configuration on a CPU, one run at a time, against 3.5–16 min on a GPU for DiffFDN.



Single training runs per configuration. Modal σ at γ = 0.9999; largest echo at a playback T60 of 2.5 s.

Figure 3. One panel per configuration. The grey markers are the gain options of Sections 4 and 5, all trained with the loss of Section 3. The ring is DiffFDN, and the blue dot the transient terms of Section 6, with free gains.

The other measures (DiffFDN / ours; modal measures in dB relative to the mean, at γ = 0.9999; NED = normalized echo density [5]; EDT at a playback T60 of 2.5 s; onset = mean level of the first 25 ms after the first echo against the tail, decay removed):

Configuration	p99 (dB)	loudest mode (dB)	NED 0.1–0.5 s	EDT (s)	onset (dB)	our training time
N4 set #1	**+6.49** / +7.11	**+7.80** / +8.57	0.16 / 0.15	3.16 / 3.02	−6.1 / −9.5	16 s
N4 set #2	+6.47 / **+5.14**	+8.06 / **+7.41**	0.30 / 0.27	2.78 / 2.86	−3.7 / −6.9	22 s
N6 set #1	**+4.81** / +5.61	**+5.68** / +7.98	0.65 / 0.69	2.97 / 2.93	−4.1 / −6.0	25 s
N6 set #2	+5.72 / **+5.35**	**+7.69** / +8.15	0.67 / 0.69	2.79 / 2.87	−2.3 / −5.2	26 s
N8 set #1	**+4.47** / +4.87	**+5.92** / +6.29	0.89 / 0.95	2.81 / 2.74	−2.8 / −5.1	34 s
N8 set #2	+6.31 / **+4.76**	+7.46 / **+6.20**	0.95 / 0.96	2.54 / 2.59	−2.2 / −5.9	86 s

Early echo density is the same within 0.06, and the late decay is identical (T60 from −5 to −35 dB at 2.48–2.50 s for both). The onset now builds up like DiffFDN's, starting further below the tail (onset column). The loudest early sample is equal on N4 set #1, 0.7 dB louder on N4 set #2 and 1.9–6.5 dB quieter on the other four.

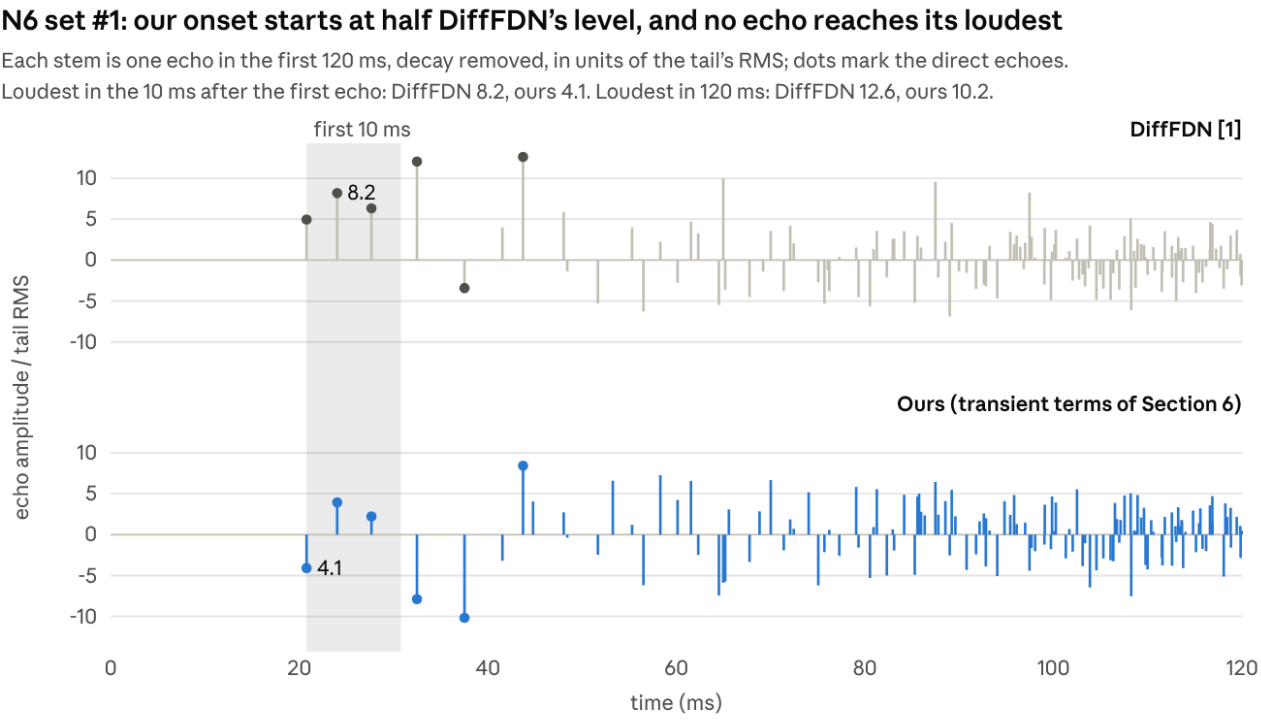


Figure 4. N6 set #1, echo by echo. DiffFDN's first three direct echoes reach 5–8 times the tail's RMS, and two later ones exceed 12. Ours start near 4, and no echo exceeds 10.2, while more mid-level echoes fill the response from about 50 ms on.

Sign balance of the onset. DiffFDN's early echoes lean to one sign: on N4 set #2 its first 13 echoes are all positive. The init of [1] gives every line's b and c the same sign at N = 4, and mostly so at N = 6 and 8, and DiffFDN's training keeps that. Measured as $|\Sigma h| / \sqrt{(\Sigma h^2)}$ over the first 100 ms (about 1 for random signs), DiffFDN's onsets score 1.3–3.4 and ours 0.1–0.8, except 2.1 on N4 set #2. The transient terms break the sign coherence as a side effect; the whole network's DC gain shows no systematic difference.

8. Limitations and next steps

These are first results, not a finished study:

- **Single runs.** Each entry is one training run, and differences of a few tenths of a dB are within run-to-run noise. The free-gain runs start from the init of [1] and are deterministic, so restarts need perturbed starting points. Keeping the best of several runs, at 16–86 s each, is the obvious next step.
- **Perceptual evaluation.** All comparisons here are objective measures. A blind listening test with the conditions of [1] would be needed to compare the methods perceptually.
- **Targets read off DiffFDN.** The build-up ramp (−6 dB over 150 ms) and the direct-echo ceiling (+10 to +22 dB) were set after measuring DiffFDN's onsets, and every weight was chosen by hand on single runs. The method is not yet independent of the reference it is compared with.
- **N = 4.** At N = 4 the transient terms cost coloration: σ rises 0.46 dB on N4 set #1, because four lines leave little freedom between the direct echoes and the modal excitation.
- **Scope.** Only the six configurations of [1] were tested, with frequency-independent decay. N = 8 uses a simpler objective than N = 4 and 6, and one objective for every N remains to be found.

The long-window limit in Section 3 also suggests a cheaper loss. The fourth moment of the residues, $\Sigma|\rho_i|^4$ / $(\Sigma|\rho_i|^2)^2$, can be computed directly from the modal decomposition without unrolling, and used as an objective together with the same counterweights.

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